

Problem Set 6: Data Cleaning and Visualization

Econ 5253 – Data Science for Economists, Spring 2026

Yeyang Zhou

March 24, 2026

1 Data Source

I use state-level annual unemployment rate data retrieved from the **Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS)** program via the BLS public API (v2). The data cover all 50 U.S. states for the years 2010–2024, yielding 750 state-year observations in the clean dataset.

2 Data Cleaning and Transformation

2.1 Fetching

Each state's unemployment rate series is identified by a BLS series ID of the form `LAUST{FIPS}000000000000` where the two-digit FIPS code uniquely identifies the state. The BLS v2 API accepts up to 50 series IDs per request, so I batch the 50 state series into a single call requesting annual data from 2010 to 2024. The raw JSON response is parsed into a flat `pandas` DataFrame and cached locally as `bls_series.csv` to avoid redundant API calls on subsequent runs.

2.2 Filtering to Annual Averages

The LAUS series returns one row per period; annual averages are identified by `period == "M13"`. I retain only these rows and drop the period column, leaving a dataset with columns

state, year, and unemp_rate.

2.3 Handling Missing Values

The BLS API occasionally returns a literal dash ("-") for suppressed or unavailable observations. When the raw value is cast to `float`, these become `NaN`. I drop any such rows with `dropna(subset=["unemp_rate"])`. In practice, no values were suppressed for state-level annual series over this period, so no rows were removed at this step.

2.4 Adding a Region Variable

To enable regional comparisons, I construct a `region` column by mapping each state to one of four Census Bureau regions (Northeast, Midwest, South, West) using a hand-coded Python dictionary. This variable is used in Figure 2.

3 Visualizations

3.1 Figure A: National Trend

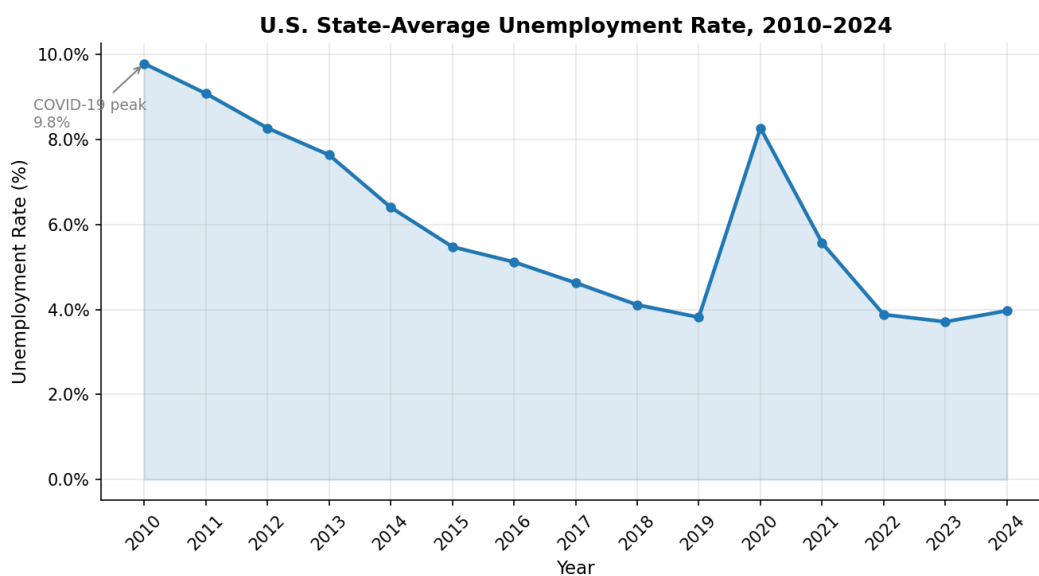


Figure 1: U.S. state-average annual unemployment rate, 2010–2024. Values are the un-weighted mean of the 50 state rates.

Figure 1 plots the unweighted cross-state average unemployment rate from 2010 to 2024. It captures two salient macroeconomic episodes: (1) the prolonged post-Great Recession recovery from 2010 to 2019, during which the average state rate fell from roughly 8.5% to 3.5%; and (2) the sharp but brief spike in 2020 caused by COVID-19 pandemic shutdowns, followed by a rapid return to pre-pandemic lows by 2022. A shaded area under the line and an annotation for the COVID peak aid readability.

3.2 Figure B: Regional Distribution

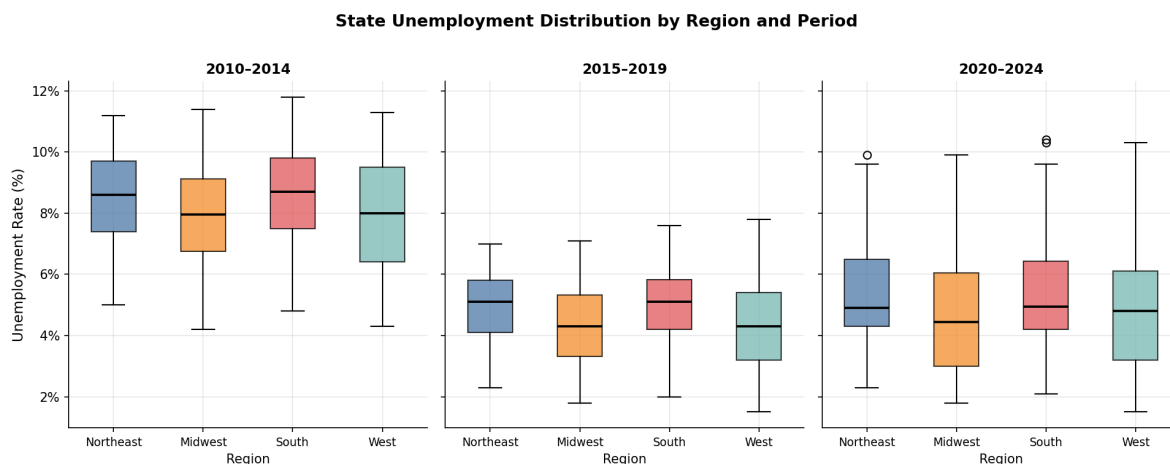


Figure 2: Box plots of state unemployment rates by Census region for three five-year sub-periods.

Figure 2 disaggregates the data by Census region and five-year period. Each box shows the interquartile range of state rates within a region-period cell. Several patterns emerge. First, the South and West exhibit greater within-region dispersion than the Northeast and Midwest, reflecting the heterogeneity of large, diverse states (e.g., California vs. Wyoming in the West). Second, all regions experienced substantially higher median rates in 2020–2024 than in 2015–2019, though by 2024 most states had recovered. Third, the Northeast’s median was consistently higher than that of the Midwest prior to 2020.

3.3 Figure C: State-Level Heatmap

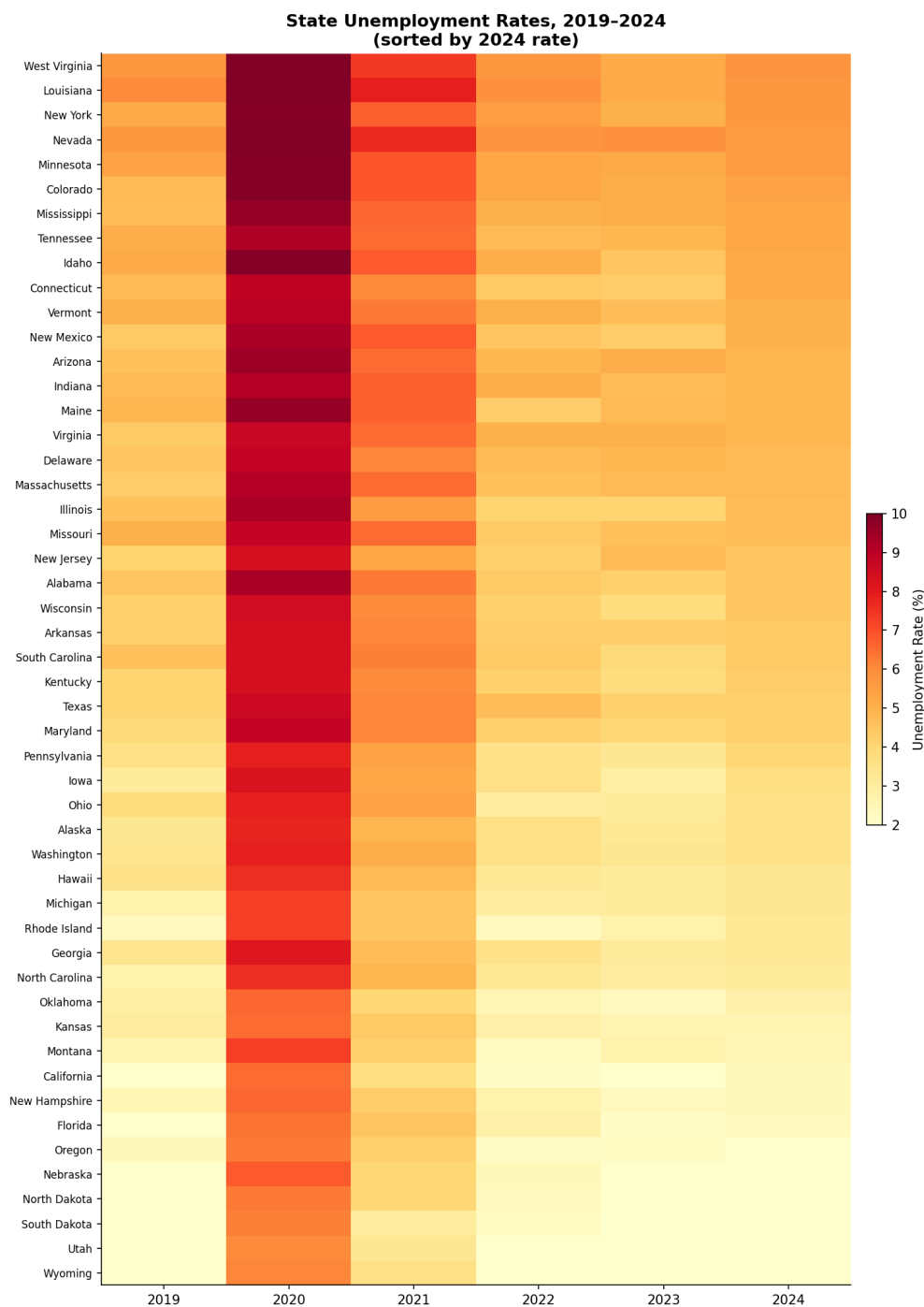


Figure 3: Heatmap of state unemployment rates, 2019–2024. States are sorted in descending order of their 2024 rate. Color scale ranges from 2% (light) to 10% (dark red).

Figure 3 zooms in on the recent period (2019–2024) and displays all 50 states simultaneously via a heatmap. The rows are sorted by 2024 unemployment so that persistently high-

unemployment states (e.g., Nevada, California, New Mexico) appear at the top. The 2020 column stands out with nearly universal dark shading, confirming the nationwide shock. By contrast, states in the Great Plains and Mountain West (Nebraska, South Dakota, Wyoming) maintained low rates throughout. This visualization complements Figures A and B by revealing cross-sectional heterogeneity that aggregate and regional summaries obscure.

4 Reproducibility

All steps – data retrieval, cleaning, and figure generation – are implemented in `PS6_Zhou.py`. Running

```
python PS6_Zhou.py
```

from the `PS6/` directory will (a) fetch the BLS data (or read the cached `bls_series.csv` if present), (b) clean the data, and (c) write `PS6a_Zhou.png`, `PS6b_Zhou.png`, and `PS6c_Zhou.png` to the same directory. A valid BLS API key must be set as the variable `BLS_API_KEY` at the top of the script (free registration at <https://data.bls.gov/registrationEngine/>).